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Centrality determination in $\sqrt{s_{\text{NN}}} = 5.02$ TeV Pb+Pb collision data in 2015

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Abstract

Supporting note for centrality analysis and recommendations in 2015 Pb+Pb data.

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1 Introduction

This note describes the geometric model for 5 TeV Pb+Pb collisions and its application to the data recorded in the 2015 LHC running. The centrality of Pb+Pb collisions is characterized by the sum of the transverse energy, ΣE_T , in both FCal modules, situated at $3.1 < |\eta| < 4.9$, similar as it was in Run 1 [1]. However, the different detector performance, running conditions, larger center-of-mass energy, and different minimum bias trigger conditions necessitate a dedicate analysis in this data. The Glauber simulation is matched to the Pb+Pb data in the region above $\Sigma E_T > 40$ GeV, which is estimated to comprise $85 \pm 1\%$ of the Glauber-like cross-section. The ΣE_T values which correspond to centile cuts, as well as the mean values of N_{part} , N_{coll} , T_{AA} and R_{coll} in different centrality ranges are presented. Uncertainties from the underlying geometric model and from the uncertainty on the efficiency are also estimated.

2 Data selection

The analysis uses the 35 Pb+Pb runs with a MinBias stream from the $\sqrt{s_{\text{NN}}} = 5.02$ TeV collision data recorded in November and December 2015. Of these, two are rejected by the GRL ¹, leaving thirty-three runs. The minimum bias ΣE_T distribution is built from the following event selection requirements:

¹ http://atlasdqm.web.cern.ch/atlasdqm/grlgen/HeavyIonP/data15_hi.periodAllYear_DetStatus-v75-repro20-01-DQDefects-00-02-02_PHYS_HeavyIonP_All_Good.xml

- logical or of the HLT_noalg_mb_L1TE50 and HLT_mb_sptrk_ion_L1ZDC_A_C_VTE50 triggers
- GRL for the given luminosity block
- presence of a reconstructed vertex
- events with characteristics consistent with that of an in-time pileup event of two Pb+Pb collisions (described below)

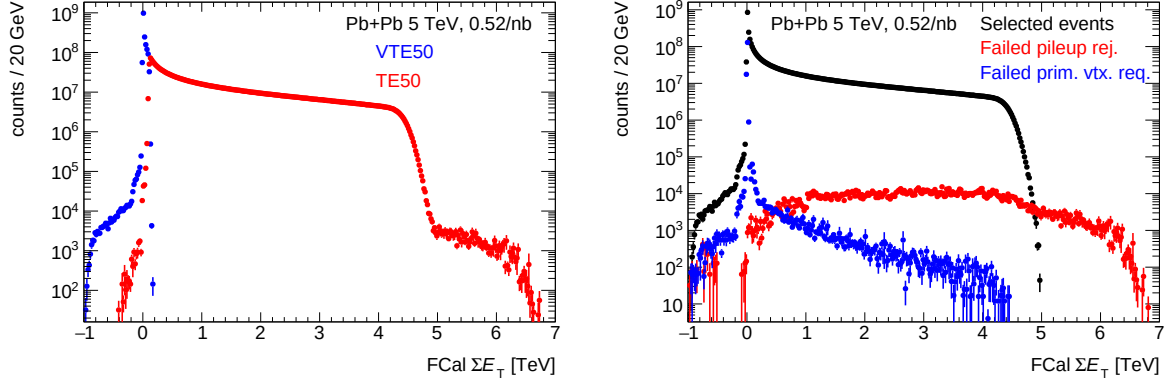


Figure 1: ΣE_T distributions in data selected by the two triggers (left), and those rejected by the vertex and pileup rejection cuts, as well as the final selected distribution (right).

The ΣE_T distribution is then built up by prescale-weighted entries from events from both triggers. Fig. 1 shows the initial distributions selected by each vertex, as well as the final distribution and that rejected by the vertex and pileup requirements.

2.1 Pileup rejection cut

The pileup rejection is implemented utilizing the anti-correlation between the energy in FCal and the number of neutrons in ZDC. The anti-correlation between the variable enables the separation of single collisions and pileup events, as the pileup events show up as a separate band in the correlation. The distribution of pileup events is estimated by convoluting the FCal energy - number of neutron distribution from data with itself and matching the scale in the region where the events are entirely from pileup. The HIPileupTool has pileup rejection cuts implemented such that at all energies no more than 0.1% of “good events” are rejected (calculated as the number of total events minus the number of estimated pileup events). The cut gives more than 80% pileup rejection for energies above 0.5 TeV and $\approx 100\%$ rejection in very central collisions, above 4.5 TeV, where the pileup fraction can be significant. More details on cuts and performance can be found in the slides here: <https://indico.cern.ch/event/559604/contributions/2257963/attachments/1315901/1971295/PileUpTool.pdf>

3 Centrality model

One million Glauber simulations of Pb+Pb collisions at 5.02 TeV were generated using the PHOBOS Glauber MC code [2] with a nucleon–nucleon cross-section of $\sigma_{NN} = 70$ mb, Woods-Saxon parameters

(R, a) = (6.62 fm, 0.546 fm), and nucleon hard core $d_N = 0.4$ fm. Events are characterized by the number of participating nucleons (N_{part}) and the number of binary nucleon–nucleon collisions (N_{coll}). Fig. 2 shows the distributions over N_{part} and N_{coll} , $P(N_{\text{part}})$ and $P(N_{\text{coll}})$ respectively.

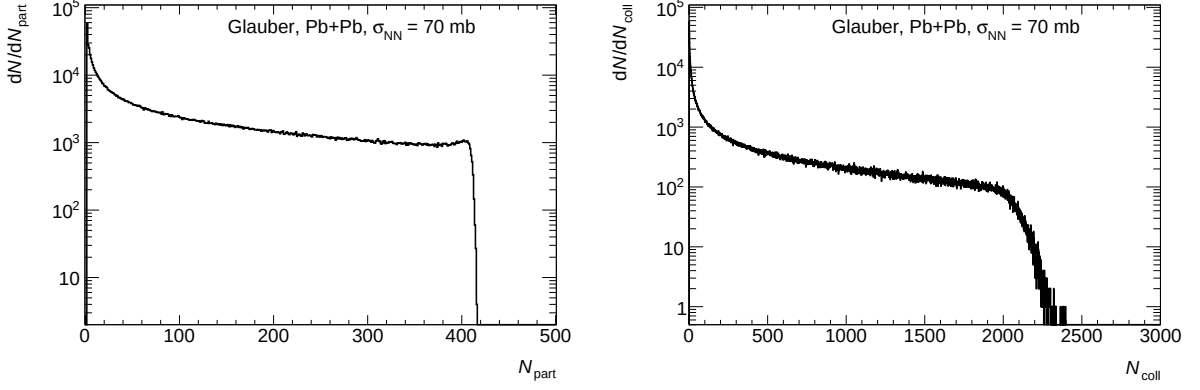


Figure 2: N_{part} (left) and N_{coll} (right) distributions in the nominal simulated Glauber event sample.

The map the distribution of geometric parameters onto the distributions in data, the so-called “two-component model” (TCM) [3] is used. The TCM hypothesizes that the overall rate of soft particle production (in this case, more specifically, the transverse energy production at large pseudorapidity ΣE_T) scales with a hybrid geometric quantity which is a linear combination of N_{part} and N_{coll} . We call this quantity the number of fundamental elements of particle production,

$$N_{\text{elem}} = x N_{\text{coll}} + (1 - x) N_{\text{part}}/2 \quad (1)$$

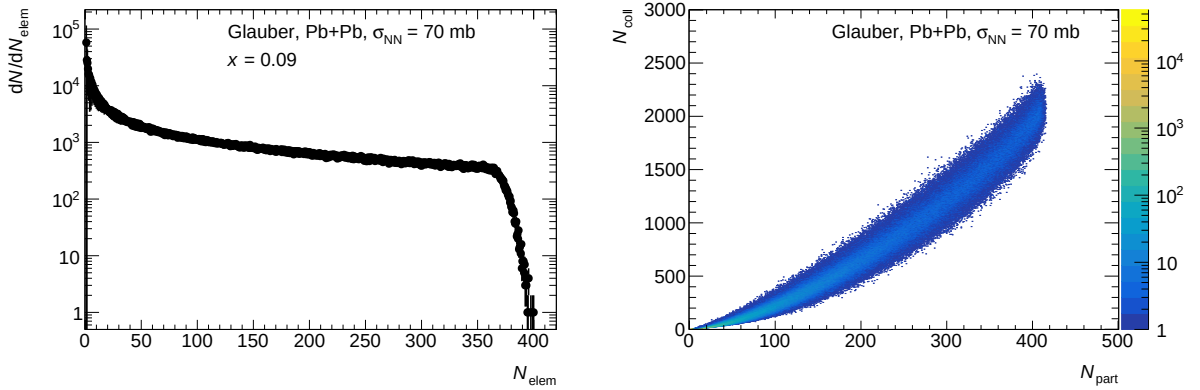


Figure 3: N_{elem} distribution with $x = 0.09$ (left) and $N_{\text{part}}-N_{\text{coll}}$ correlation (right) in the nominal simulated Glauber event sample.

where the variable x controls the linear weighting. For pp collisions, where $N_{\text{part}} = 2$ and $N_{\text{coll}} = 1$, $N_{\text{elem}} = 1$ for all values of x . Thus, schematically, the ΣE_T produced per element is equivalent to that in a single pp collision at the same $\sqrt{s}$. Fig. 3 shows the $N_{\text{part}}-N_{\text{coll}}$ correlation as well as the N_{elem} distribution with $x = 0.09$.

The ΣE_T distribution from a single element of particle production is modeled in two ways, either as a Gaussian (the simpler case, used as the default) or a Gamma distribution (used as an alternate choice in the modeling):

$$P(\Sigma E_T; N_{\text{elem}} = 1) = \text{Gaus}(\Sigma E_T; \sigma_{pp}, \mu_{pp}), \text{ or} \quad (2)$$

$$= \text{Gamma}(\Sigma E_T; \theta_{pp}, k_{pp}) \quad (3)$$

Each element is treated as producing particles independently. Thus, the ΣE_T distribution from an event with $N_{\text{elem}} > 1$ is an N_{elem} -fold convolution over the original distribution.

$$P(\Sigma E_T; N_{\text{elem}}) = \text{Gaus}(\Sigma E_T; \sigma_{pp} \sqrt{N_{\text{elem}}}, \mu_{pp} N_{\text{elem}}), \text{ or} \quad (4)$$

$$= \text{Gamma}(\Sigma E_T; \theta_{pp}, k_{pp} N_{\text{elem}}) \quad (5)$$

Finally, the ΣE_T distribution over all Pb+Pb events is given by the linear combination of the ΣE_T distribution from each N_{elem} value, weighted by the probability of that value of N_{elem} in Pb+Pb events,

$$P(\Sigma E_T) = \sum_{N_{\text{elem}}} P(\Sigma E_T; N_{\text{elem}}) P(N_{\text{elem}}) \quad (6)$$

This is the hypothetical distribution given by the centrality model. It is fit to the measured distribution in data, with the Gamma or Gaussian parameters for $N_{\text{elem}} = 1$ (the pp -like distributions) and x as free parameters.

4 Analysis

This section describes the analysis steps and application of the model to the data.

4.1 Gap analysis and working point determination

The Glauber model simulations of Pb+Pb collisions contain only the hadronic fraction of the Pb+Pb cross-section (i.e. the events where at least one pair of nucleons, one from each nucleus, interacted hadronically over a short distance scale). On the other hand, the Pb+Pb collision data contains an admixture of other events, such as photonuclear and diffractive interactions. Thus, the model will not be able to describe, even in principle, the contribution of these events. Fortunately, the non-hadronic fraction of the cross-section is characterized by small ΣE_T values. Thus, to avoid the issues associated with identifying and separating these events from the Glauber-like events, the analysis is performed above a ΣE_T *working point*, a minimum value above which the Pb+Pb are all Glauber-like.

The working point value is determined via a reconstructed rapidity gap analysis, similar to that performed for the Run 1 p +Pb centrality analysis [4] which itself was based on a measurement of pseudorapidity gap

cross-sections in 7 TeV pp collisions [5]. Glauber-like Pb+Pb events are characterized by small rapidity gaps – consecutive regions of rapidity without particle activity – while diffractive and photonuclear events both preferentially have a large activity gap on one side of the detector. To test for the presence of events with large rapidity gaps at a given value of ΣE_T , the edge gap $\Delta\eta_{\text{edge}}$ is determined for each Pb+Pb event in data.

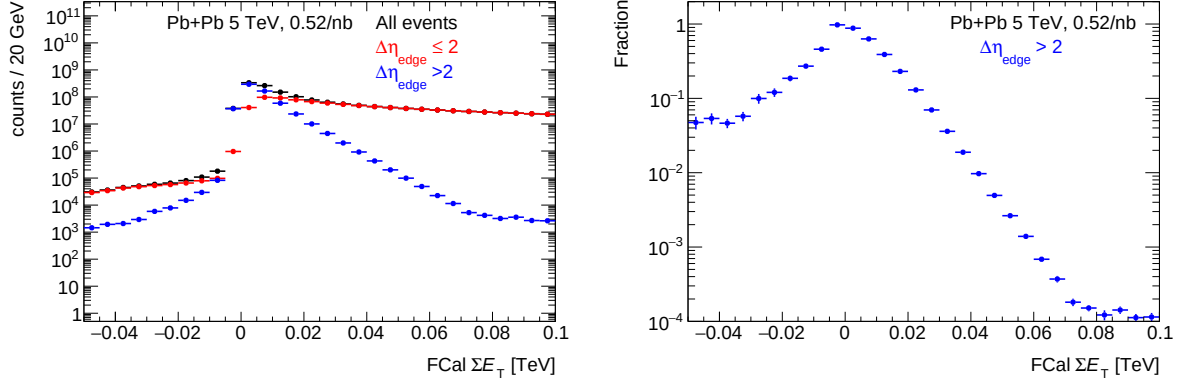


Figure 4: ΣE_T distributions for events that pass and fail the edge gap requirement (left), and the fraction of events that fail as a function of ΣE_T (right).

The pseudorapidity gaps are constructed by examining the 49 pseudorapidity rings of $\delta\eta = 0.2$ between $\eta = \pm 4.9$. A pseudorapidity ring is considered empty if it has no clusters with central cell above some minimum significance value ($S = E/\sigma_{\text{noise}}$) or minimum E_T value, where the minimum significance and E_T values are calibration by comparing distributions of cluster energies from empty and paired bunches. The pseudorapidity edge gap $\Delta\eta_{\text{edge}}$ is defined as the larger of the number of sequential empty rings, starting either from $\eta = +4.9$ or $\eta = -4.9$ and events are considered as diffractive candidates if $\Delta\eta_{\text{edge}} > 2.0$. It is found that for events with $\Sigma E_T > 40$ GeV, the fraction of events with these large values of $\Delta\eta_{\text{edge}}$ is negligible. Thus, 40 GeV is chosen as the working point. Fig. 4 shows the ΣE_T distributions for events which pass and fail this cut, and the fraction that fail as a function of ΣE_T . It can be seen that above 40 GeV, 1% of fewer of the Pb+Pb events in data fail this cut, demonstrating that they are dominated by the Glauber-like Pb+Pb cross-section.

4.2 Choice of x and fit to data

As in the Run 1 analysis [1], the plausible range of x values was set using external constraints as input. Two potential values of x were determined through different approaches.

First, the ΣE_T distribution in pp 5 TeV data can be used to constrain x . Since the average ΣE_T is linear in N_{elem} , and $N_{\text{elem}} = 1$ represents the “ pp -like” level of particle production, the average ΣE_T in Pb+Pb collisions can be related to the level in pp collisions through the average value of N_{elem} as follows,

$$\langle \Sigma E_T \rangle^{\text{Pb+Pb}} = \langle N_{\text{elem}} \rangle \langle \Sigma E_T \rangle^{pp} \quad (7)$$

$$\langle \Sigma E_T \rangle^{\text{Pb+Pb}} = \left(x \langle N_{\text{coll}} \rangle + (1-x) \langle N_{\text{part}} \rangle / 2 \right) \langle \Sigma E_T \rangle^{pp} \quad (8)$$

$$\langle \Sigma E_T \rangle^{\text{Pb+Pb}} / \langle \Sigma E_T \rangle^{pp} = x \left(\langle N_{\text{coll}} \rangle - \langle N_{\text{part}} \rangle / 2 \right) + \langle N_{\text{part}} \rangle / 2 \quad (9)$$

$$\left(\langle \Sigma E_T \rangle^{\text{Pb+Pb}} / \langle \Sigma E_T \rangle^{pp} - \langle N_{\text{part}} \rangle / 2 \right) / \left(\langle N_{\text{coll}} \rangle - \langle N_{\text{part}} \rangle / 2 \right) = x \quad (10)$$

In Eq. 10, the quantities on the left side are either estimates in data or given by the geometric model. Unfortunately, because of the potential contamination of the Pb+Pb ΣE_T distribution by non-Glauber-like events, and because of the difficulty of obtaining a ΣE_T distribution in pp collisions which is not even slightly biased by a trigger selection, this cannot be done with perfect precision. Nevertheless, an analysis performed of this type gives a value of $x \approx 0.11$.

Second, changing the value of x was found to be the major factor which affected the shape of N_{elem} distribution and, therefore, the overall shape of the ΣE_T distribution in data. Thus, in an alternate analysis, x was varied in fine steps, and the Gaussian or Gamma parameters allowed to flow, until the ratio of the simulated to the measured distribution was as close to flat as possible in the shoulder region. The value of x determined by this procedure was ≈ 0.07 .

Thus, the nominal value of x was chosen to lie between these two, with the uncertainty in the parameter covering the range, $x = 0.09 \pm 0.02$. By comparison, the value of x determined in the Run 1 analysis through different methods than the one described here was $x = 0.088$.

The nominal fit to the data therefore fixes $x = 0.09$ and allows the Gaussian parameters μ, σ for the pp -like ΣE_T distribution to float. Furthermore, the fit is performed only in the region above the working point $\Sigma E_T > 40$ GeV, and the data and simulated distributions are normalized to have the same integral above this value. Fig 5 shows the nominal (using a Gaussian response) fit and its parameters, and simulated/data comparison, as well the result of a fit with a Gamma response.

The nominal fit finds $\mu = 11.8$ GeV, $\sigma = 5.65$ GeV for the pp -like ΣE_T distribution, and an efficiency (fraction of the Glauber-like distribution) above the working point of 84.9%, which was rounded up to 85% in the analysis.

The fits were also performed holding x to be 0.07 (from the shape analysis) or 0.11 (from the pp -scaling analysis), with the results shown in Fig. 6. It can be seen that these changes in x induce a change in the efficiency by $\pm 1\%$. This finding will be used later to test the sensitivity of the results to this uncertainty in the efficiency.

4.3 Determination of ΣE_T cuts

The ΣE_T values which correspond to the centile positions were determined from the data. The minimum bias Pb+Pb events above the working point were sorted by their ΣE_T values. Since the nominal fit finds an efficiency of 85%, the events were divided into 85 bins of equal sizes, and the dividing values were taken to be the positions of the centile cuts. The full results for these are given in Section 6.1.

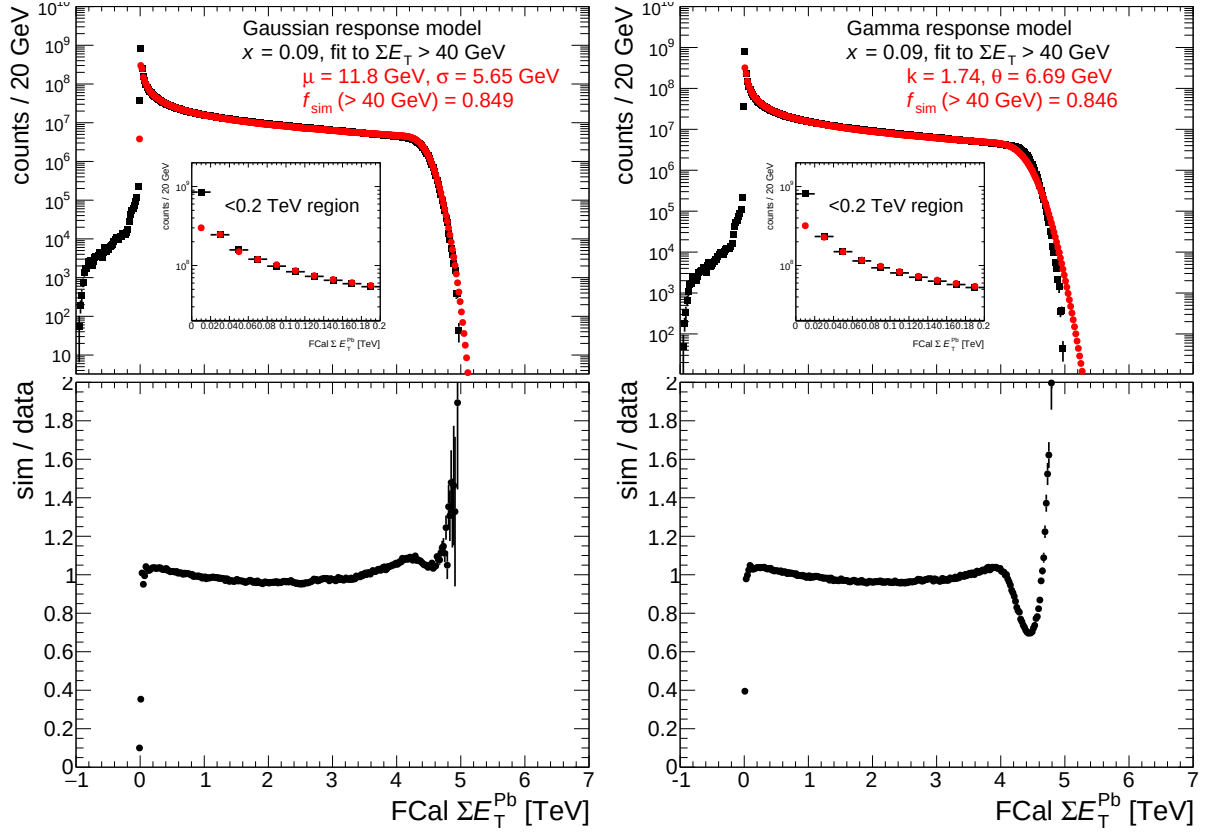


Figure 5: Fit result showing a Glauber + Gaussian fit (left, the default model) and Glauber + Gamma fit (right, an alternate modeling) to the data in the region $\Sigma E_T > 40$ GeV, fixing $x = 0.09$.

4.4 Extraction of geometric quantities

The geometric quantities are determined by examining the distribution of N_{part} and N_{coll} values in the centile bins in the simulated distribution. That is, after the optimal fit is found, the two-dimensional $(\Sigma E_T, N_{\text{part}})$ and $(\Sigma E_T, N_{\text{coll}})$ distributions are populated using the best fit parameters, and the mean values of N_{part} and N_{coll} in each ΣE_T range can be determined. The mean value of T_{AA} is determined as the ratio of the mean N_{coll} to σ_{NN} , which is typically 70 mb except in two of the Glauber variations. The full results for the geometric quantities are given in Section 6.1.

5 Systematic uncertainties

For this analysis, it was found that two major sources of systematic uncertainties were important. For each geometric parameter (N_{part} , N_{coll} , T_{AA}), alternate values were determined by considering the variations in the procedure below. The difference in the result from each pair of variations was symmetrized, and half of the difference between the up/down variations was assigned as the systematic uncertainty from that source. Then, the uncertainties from each source were added in quadrature.

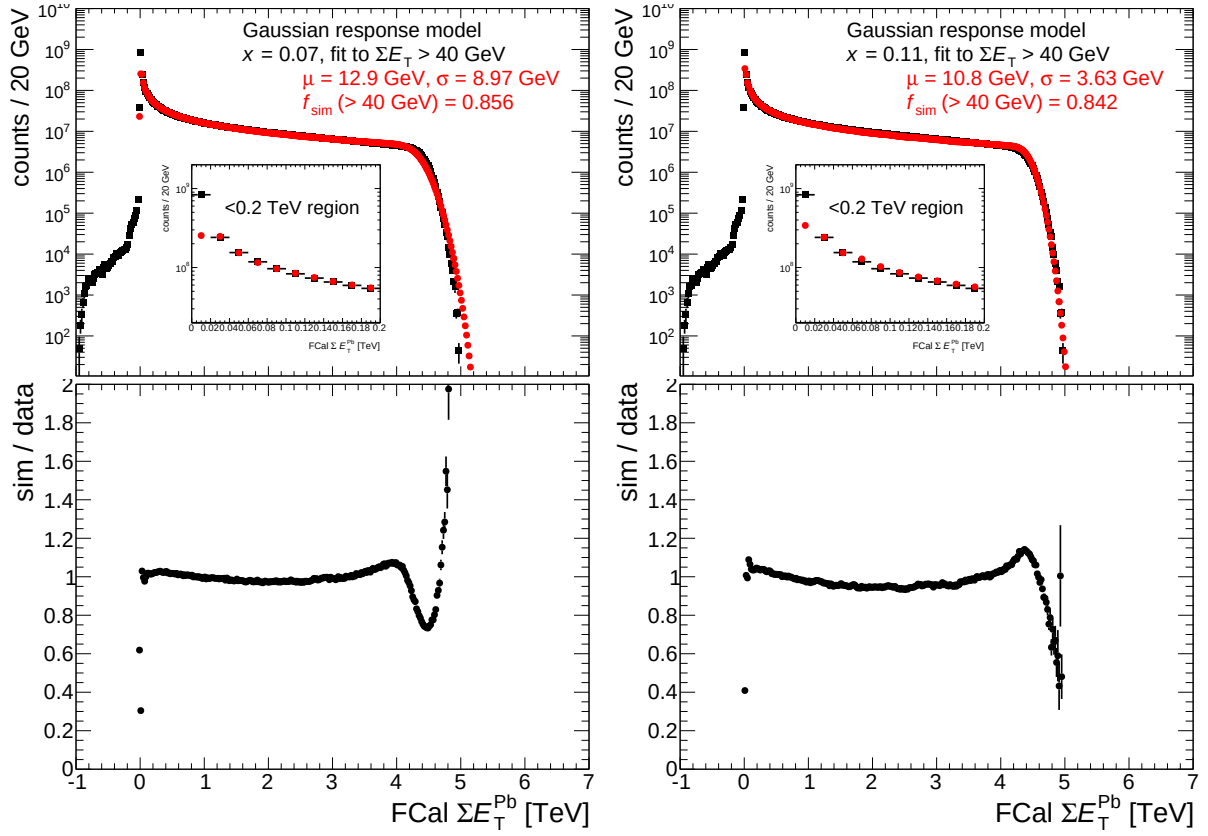


Figure 6: Fit result showing Glauber + Gaussian fits to the data in the region $\Sigma E_T > 40$ GeV, fixing $x = 0.07$ (left) and $x = 0.11$ (right).

Figure 7 shows the effects of the variations described below on the mean values of N_{part} , N_{coll} and T_{AA} in 10%-wide centrality bins.

5.1 Geometric uncertainties

To determine the uncertainties from the geometric Glauber modeling, six alternate sets of Glauber events were generated, with the following three pairs of up/down variations from those used to generate the nominal set of events:

1. $\sigma_{\text{NN}} = 65$ mb
2. $\sigma_{\text{NN}} = 75$ mb
3. Woods-Saxon parameters $(R, a) = (6.68 \text{ fm}, 0.536 \text{ fm})$
4. Woods-Saxon parameters $(R, a) = (6.56 \text{ fm}, 0.556 \text{ fm})$
5. Nucleon hard core $d_N = 0.2$ fm
6. Nucleon hard core $d_N = 0.6$ fm

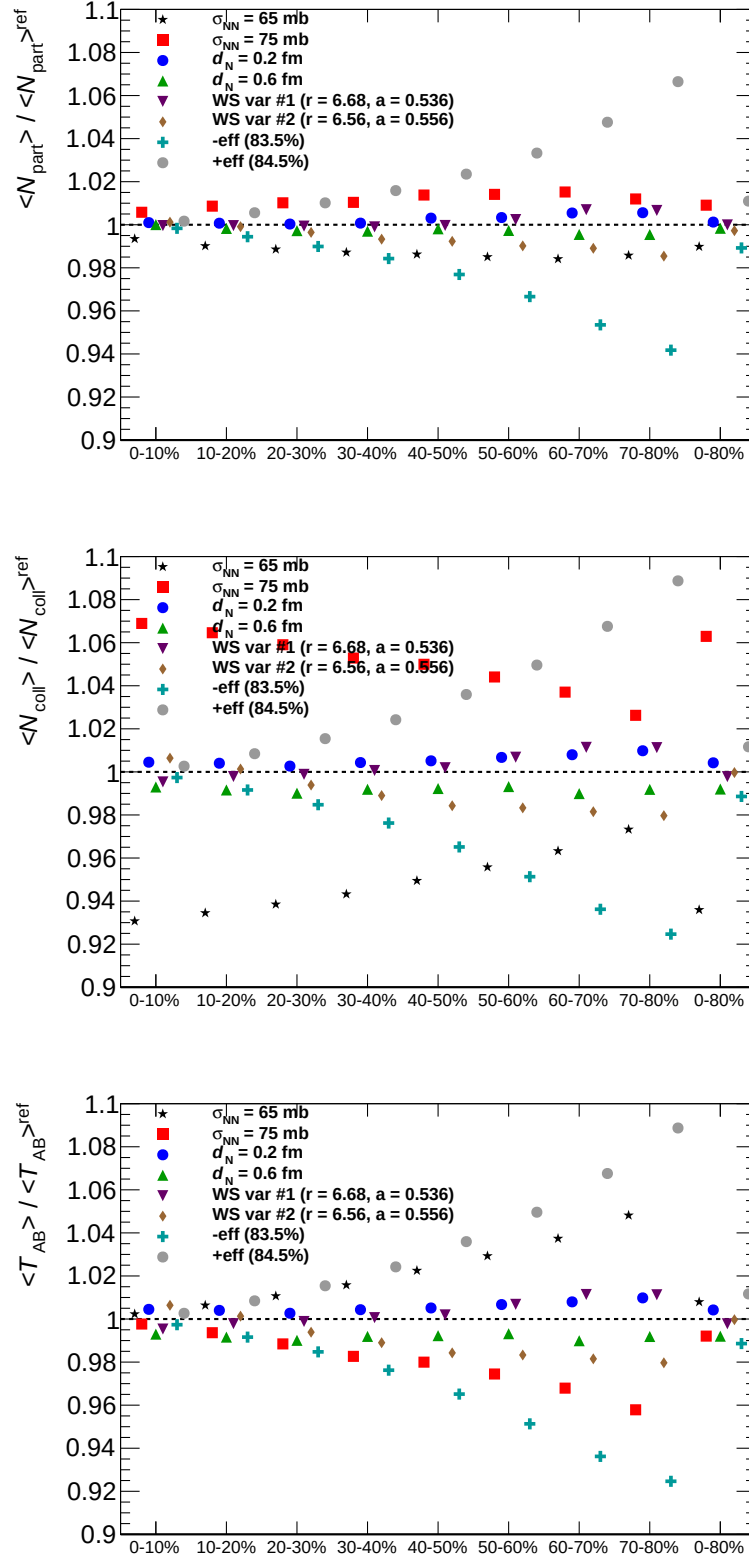


Figure 7: Ratios of the geometric quantities extracted under systematic variations to the nominal ones, shown for 10%-sized centrality bins.

The fit to the data was repeated using the alternate N_{elem} distribution provided by these versions of the model, and determining the alternate version of geometric quantities.

5.2 Efficiency uncertainty

To determine the uncertainty in the extracted results from the uncertainty in the efficiency, the geometric parameters were extracted from the simulated distribution under the assumption that 84% or 86% of the Glauber distribution lies above the working point, instead of 85%. In that case, a given centrality range would actually correspond to a range where the values are multiplied by 86/85 or 84/85, respectively. The geometric parameters in these modified bins were used as the alternate results from this systematic uncertainty.

Additionally, for physics analyses which do not use the geometric parameters (for example, correlation analyses instead of analyses which measure hard-process rates), an alternate version of the ΣE_T cuts was derived in which the minimum bias distribution in data was divided into 84 or 86 bins instead of the nominal 85.

5.3 Other uncertainties

Other potential sources of systematic uncertainty were evaluated, but were found to have a negligible effect with respect to the two main categories listed above.

The sensitivity of the choice of Gaussian vs. Gamma distribution in the fit model was tested. The Gaussian distributions gave a fit result which agreed better with the data, but the extracted efficiency and geometric quantities were negligibly different.

The sensitivity to the working point value above which the fit was performed was also examined, by raising it to 60 GeV or down to 30 GeV (where the admixture of non-Glauber-like Pb+Pb events rises but does not dominate the distribution), and the results were found to be consistent.

6 Results

6.1 Nominal and variant ΣE_T cuts

Table 1 shows the ΣE_T cuts for 1%-wide centile boundaries, for the nominal (85% efficiency) analysis, as well as the alternate 84% and 86% definitions.

6.2 Flatness and run stability checks

To validate the ΣE_T cuts derived above, the relative fraction of minimum bias events passing event selection cuts in each centrality bin were examined to ensure that they give a flat distribution in centrality. Additionally, this flatness was tested in individual Pb+Pb runs to make sure that a given centrality cut was making a consistent selection throughout the data set.

The number of events in each centrality bin (1%-wide and also 10%-wide centiles) was counted in minimum bias data in each of the 35 Pb+Pb runs. Then, the number of events was divided by the total number of events with $\Sigma E_T > 40$ GeV in the run, and multiplied by the efficiency above the working point (0.85). Thus, if the cuts are working correctly, according to this definition a centrality range that is N centiles wide should have a relative fraction of $0.01N$.

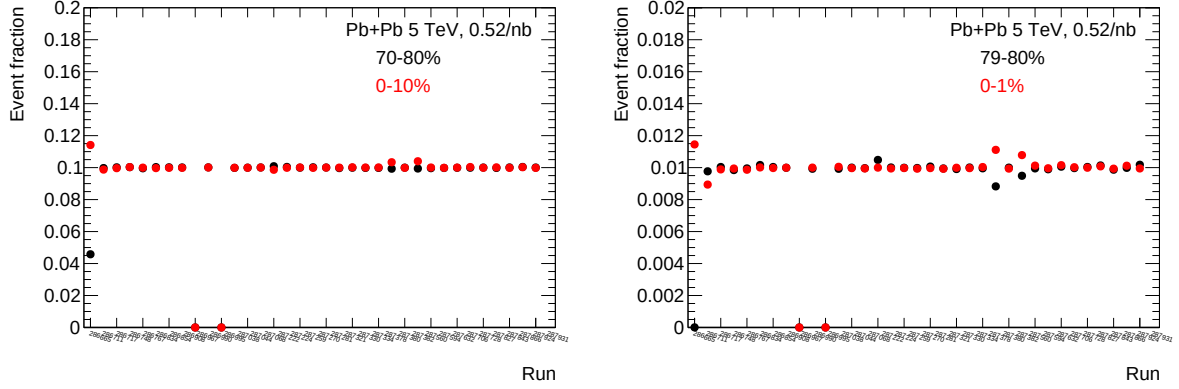


Figure 8: Fraction of events in the given centrality bin, shown as a function of Pb+Pb run number. The most central and most peripheral selections are evaluated in 10% wide bins (left) or 1% wide bins (right).

Figure 8 shows the relative fraction as a function of run number for the most central and most peripheral event selection in 10%-wide and 1%-wide bins. For most cases, the event fractions are consistent with the size of the centrality bin. However, there are a few outliers:

- The first run, #286665, included the TE50 minimum bias trigger in its menu, but not the VTE50 trigger. Thus, the ΣE_T distribution has the correct shape for centralities in the range 0 to $\approx 75\%$, but is missing events in the range $\approx 75\%$ to 80%. It may thus still be usable for analyses, depending on the particular trigger selection.
- There are two runs which are shown as having an event fraction of zero (#286967 and #286995). This is a plotting artifact as both runs are actually excluded from the Pb+Pb GRL. However, if the GRL cut is turned off, the ΣE_T distribution still resembles like that for events which pass the cuts. Thus, it may be possible, at least from the perspective of the centrality analysis, to still use these runs.
- Two other runs (#286380 and #286560) have a slightly increased and decreased event fraction in central and peripheral events, respectively. That is, the ΣE_T distribution in these runs has a slightly different shape from that in the others. Notably, these runs have a very small luminosity compared to the adjacent runs. Thus, they may be used with this caveat.

6.3 Geometric quantities

Tables 2, 3 and 4 show the full set of results for N_{part} , N_{coll} , and T_{AA} respectively. Table 5 shows the results for R_{coll} for some selections of the peripheral (denominator) bin.

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Centile	ΣE_T [TeV]		
	Nominal (85%)	Down Variation 84%	Up Variation 86%
80 %	0.063719	0.058527	0.069136
79 %	0.06956	0.064057	0.07531
78 %	0.075838	0.069999	0.081906
77 %	0.082548	0.076379	0.088954
76 %	0.089723	0.083214	0.09648
75 %	0.097388	0.09052	0.104536
74 %	0.105619	0.098342	0.113101
73 %	0.114352	0.106732	0.122216
72 %	0.123657	0.115651	0.131908
71 %	0.133573	0.12515	0.142242
70 %	0.14414	0.135286	0.15318
69 %	0.155307	0.146078	0.164806
68 %	0.167193	0.157514	0.177083
67 %	0.179776	0.169673	0.190096
66 %	0.193096	0.182579	0.203824
65 %	0.207148	0.196214	0.218281
64 %	0.22199	0.210613	0.233553
63 %	0.237615	0.225837	0.249597
62 %	0.25407	0.241851	0.266458
61 %	0.27137	0.258713	0.284192
60 %	0.289595	0.276461	0.302826
59 %	0.308686	0.295168	0.322382
58 %	0.328744	0.314811	0.342787
57 %	0.349697	0.335361	0.364095
56 %	0.371561	0.356885	0.386397
55 %	0.394518	0.379373	0.409774
54 %	0.418573	0.402984	0.434162
53 %	0.443549	0.427675	0.459497
52 %	0.46959	0.453331	0.485844
51 %	0.49675	0.480118	0.513454
50 %	0.525092	0.508105	0.542107
49 %	0.554569	0.537254	0.571956
48 %	0.585275	0.567621	0.602869
47 %	0.617108	0.599129	0.634998
46 %	0.65018	0.631882	0.668303
45 %	0.684377	0.665879	0.702811
44 %	0.719896	0.701144	0.738626
43 %	0.756791	0.737746	0.775749
42 %	0.795018	0.775749	0.814217
41 %	0.834538	0.815154	0.85383
40 %	0.87541	0.855791	0.894902
39 %	0.917795	0.897878	0.937428
38 %	0.961609	0.941579	0.981313
37 %	1.0068	0.986655	1.02683
36 %	1.05367	1.0334	1.07376
35 %	1.10211	1.08171	1.12226
34 %	1.15214	1.13168	1.17244
33 %	1.20373	1.18333	1.22398
32 %	1.25693	1.23644	1.27719
31 %	1.31197	1.29143	1.33218
30 %	1.36875	1.34812	1.38892
29 %	1.42719	1.40677	1.44736
28 %	1.48744	1.46718	1.50764
27 %	1.55005	1.52974	1.57
26 %	1.61434	1.59422	1.63416
25 %	1.68058	1.66063	1.70029
24 %	1.74932	1.7295	1.76876
23 %	1.81997	1.80042	1.83945
22 %	1.89316	1.87379	1.91222
21 %	1.96859	1.94949	1.98743
20 %	2.04651	2.0277	2.06495
19 %	2.12711	2.10858	2.14525
18 %	2.21002	2.19205	2.22772
17 %	2.29572	2.27819	2.31305
16 %	2.38468	2.36745	2.40148
15 %	2.47658	2.45993	2.49292
14 %	2.57162	2.55558	2.58742
13 %	2.66999	2.65451	2.68513
12 %	2.77237	2.75761	2.78688
11 %	2.87864	2.86439	2.89266
10 %	2.98931	2.9759	3.00247
9 %	3.10407	3.09149	3.11632
8 %	3.22397	3.21218	3.23537
7 %	3.34945	3.33876	3.35982
6 %	3.48077	3.47119	3.49022
5 %	3.61844	3.60999	3.62674
4 %	3.7635	3.75643	3.77047
3 %	3.91763	3.91194	3.92323
2 %	4.08137	4.07724	4.08541
1 %	4.26258	4.26017	4.2649

Table 1: Overview of ΣE_T cut results, showing the cuts in 1%-sized bins for the nominal and variant efficiency cases.

range	N_{part}	δN_{part}	$\delta N_{\text{part}}/N_{\text{part}}$
1%-wide bins (for < 10%)			
0 - 1%	402.933	1.51967	0.00377151
1 - 2%	395.072	1.76474	0.00446688
2 - 3%	385.43	1.93715	0.00502594
3 - 4%	374.886	2.13641	0.00569882
4 - 5%	364.15	2.30819	0.00633856
5 - 6%	353.66	2.46451	0.00696858
6 - 7%	343.259	2.60816	0.00759823
7 - 8%	332.954	2.72044	0.00817061
8 - 9%	322.768	2.86742	0.00888385
9 - 10%	312.917	2.95846	0.00945446
5%-wide bins			
0 - 5%	384.494	1.92651	0.00501049
5 - 10%	333.112	2.71855	0.00816109
10 - 15%	285.244	2.87078	0.0100643
15 - 20%	242.921	2.88023	0.0118567
20 - 25%	205.637	2.87468	0.0139794
25 - 30%	172.808	2.78709	0.0161282
30 - 35%	144.107	2.68282	0.0186168
35 - 40%	118.754	2.55253	0.0214942
40 - 45%	96.6306	2.45894	0.0254469
45 - 50%	77.4079	2.31635	0.029924
50 - 55%	60.929	2.10479	0.034545
55 - 60%	46.9523	1.88754	0.0402013
60 - 65%	35.2932	1.6779	0.0475418
65 - 70%	25.8426	1.4168	0.0548243
70 - 75%	18.2855	1.2222	0.0668397
75 - 80%	12.5347	0.784379	0.0625766
10%-wide bins			
0 - 10%	358.803	2.31243	0.00644485
10 - 20%	264.082	2.86574	0.0108517
20 - 30%	189.223	2.82774	0.014944
30 - 40%	131.431	2.61435	0.0198914
40 - 50%	87.0192	2.38637	0.0274235
50 - 60%	53.9407	1.99509	0.0369868
60 - 70%	30.5679	1.54673	0.0506
70 - 80%	15.4101	0.998343	0.0647849
20%-wide bins			
0 - 20%	311.443	2.55711	0.00821052
20 - 40%	160.327	2.70923	0.0168982
40 - 60%	70.4799	2.18649	0.0310229
60 - 80%	22.989	1.27062	0.0552709
40%-wide bins			
0 - 40%	235.885	2.56666	0.010881
40 - 80%	46.7345	1.72048	0.0368139
80%-wide bins			
0 - 80%	141.31	2.06921	0.0146431
Additional bins			
30 - 50%	109.225	2.49724	0.0228633
50 - 80%	33.3062	1.50877	0.0453

Table 2: Overview of N_{part} results, showing the central value and the absolute and relative uncertainty in bins of different sizes.

range	N_{coll}	δN_{coll}	$\delta N_{\text{coll}}/N_{\text{coll}}$
1%-wide bins (for < 10%)			
0 - 1%	2038.21	149.196	0.0731997
1 - 2%	1923.23	136.402	0.0709237
2 - 3%	1827.99	128.066	0.0700585
3 - 4%	1738.4	121.036	0.0696249
4 - 5%	1654.04	114.022	0.0689358
5 - 6%	1575.75	107.954	0.0685096
6 - 7%	1501.24	102.858	0.0685151
7 - 8%	1429.39	98.0046	0.0685638
8 - 9%	1360.92	92.8471	0.0682237
9 - 10%	1297.42	87.9491	0.0677877
5%-wide bins			
0 - 5%	1836.37	129.733	0.0706465
5 - 10%	1432.95	97.9097	0.0683275
10 - 15%	1125.79	75.0048	0.0666242
15 - 20%	881.041	57.3416	0.0650839
20 - 25%	684.169	43.4822	0.0635548
25 - 30%	525.212	32.2313	0.0613683
30 - 35%	397.035	24.2742	0.0611387
35 - 40%	295.42	17.633	0.0596878
40 - 45%	215.435	13.2816	0.0616502
45 - 50%	153.374	9.77774	0.0637511
50 - 55%	106.819	7.05961	0.0660898
55 - 60%	72.535	5.05648	0.0697109
60 - 65%	48.0477	3.6175	0.0752899
65 - 70%	31.0363	2.508	0.0808086
70 - 75%	19.3834	1.78473	0.0920749
75 - 80%	11.8508	0.977521	0.0824859
10%-wide bins			
0 - 10%	1634.66	113.795	0.0696139
10 - 20%	1003.42	66.1424	0.0659173
20 - 30%	604.691	37.8282	0.0625579
30 - 40%	346.228	20.9222	0.0604291
40 - 50%	184.404	11.5119	0.0624276
50 - 60%	89.6768	6.04788	0.0674409
60 - 70%	39.542	3.05763	0.0773262
70 - 80%	15.6171	1.37579	0.0880948
20%-wide bins			
0 - 20%	1319.04	89.8533	0.0681204
20 - 40%	475.459	29.2761	0.0615743
40 - 60%	137.041	8.73043	0.0637069
60 - 80%	27.5795	2.20764	0.0800464
40%-wide bins			
0 - 40%	897.248	59.1814	0.0659588
40 - 80%	82.3101	5.39084	0.0654942
80%-wide bins			
0 - 80%	489.779	31.7476	0.0648203
Additional bins			
30 - 50%	265.316	16.14	0.0608331
50 - 80%	48.2786	3.45521	0.0715681

Table 3: Overview of N_{coll} results, showing the central value and the absolute and relative uncertainty in bins of different sizes.

range	T_{AA} [mb ⁻¹]	δT_{AA} [mb ⁻¹]	$\delta T_{AA}/T_{AA}$
1%-wide bins (for < 10%)			
0 - 1%	29.1173	0.268315	0.00921499
1 - 2%	27.4747	0.224509	0.0081715
2 - 3%	26.1141	0.217947	0.00834594
3 - 4%	24.8342	0.208764	0.00840628
4 - 5%	23.6291	0.207412	0.00877783
5 - 6%	22.5107	0.20428	0.00907477
6 - 7%	21.4463	0.205638	0.00958849
7 - 8%	20.4199	0.202058	0.00989516
8 - 9%	19.4417	0.188055	0.00967274
9 - 10%	18.5346	0.182118	0.00982586
5%-wide bins			
0 - 5%	26.2339	0.220575	0.00840801
5 - 10%	20.4707	0.195109	0.00953116
10 - 15%	16.0827	0.18419	0.0114527
15 - 20%	12.5863	0.174254	0.0138447
20 - 25%	9.77385	0.175761	0.0179828
25 - 30%	7.50302	0.172365	0.0229728
30 - 35%	5.67193	0.155819	0.027472
35 - 40%	4.22029	0.144595	0.0342619
40 - 45%	3.07765	0.122537	0.0398153
45 - 50%	2.19105	0.10262	0.0468362
50 - 55%	1.52598	0.0829883	0.0543836
55 - 60%	1.03621	0.0653272	0.0630441
60 - 65%	0.686395	0.049544	0.0721799
65 - 70%	0.443376	0.0367004	0.082775
70 - 75%	0.276906	0.0277697	0.100286
75 - 80%	0.169297	0.0149006	0.0880144
10%-wide bins			
0 - 10%	23.3523	0.203451	0.00871226
10 - 20%	14.3345	0.177361	0.012373
20 - 30%	8.63844	0.173873	0.0201278
30 - 40%	4.94611	0.150201	0.0303674
40 - 50%	2.63435	0.112539	0.0427197
50 - 60%	1.2811	0.0741528	0.0578823
60 - 70%	0.564885	0.0431211	0.076336
70 - 80%	0.223102	0.0212812	0.095388
20%-wide bins			
0 - 20%	18.8434	0.178888	0.00949343
20 - 40%	6.79227	0.161691	0.0238051
40 - 60%	1.95772	0.093322	0.0476686
60 - 80%	0.393994	0.0321946	0.0817136
40%-wide bins			
0 - 40%	12.8178	0.157846	0.0123145
40 - 80%	1.17586	0.0627438	0.05336
80%-wide bins			
0 - 80%	6.99684	0.107036	0.0152978
Additional bins			
30 - 50%	3.79023	0.131259	0.034631
50 - 80%	0.689695	0.0461771	0.066953

Table 4: Overview of T_{AA} results, showing the central value and the absolute and relative uncertainty in bins of different sizes.

range	R_{coll}	δR_{coll}	$\delta R_{\text{coll}}/R_{\text{coll}}$
10%-wide bins / 60-80%			
0 - 10% / 60-80%	59.2707	4.68575	0.0790568
10 - 20% / 60-80%	36.3826	2.59469	0.0713167
20 - 30% / 60-80%	21.9253	1.36483	0.0622488
30 - 40% / 60-80%	12.5538	0.648621	0.0516673
40 - 50% / 60-80%	6.68628	0.261014	0.0390372
50 - 60% / 60-80%	3.25157	0.077168	0.0237326
20%-wide bins / 60-80%			
0 - 20% / 60-80%	47.8266	3.63987	0.0761056
20 - 40% / 60-80%	17.2396	1.00659	0.0583886
40 - 60% / 60-80%	4.96892	0.169071	0.0340256
20%-wide bins / 40-80%			
0 - 20% / 40-80%	16.0252	0.770129	0.0480573
20 - 40% / 40-80%	5.77644	0.174115	0.0301423

Table 5: Overview of R_{coll} results, showing the central value and the absolute and relative uncertainty in bins of different sizes.